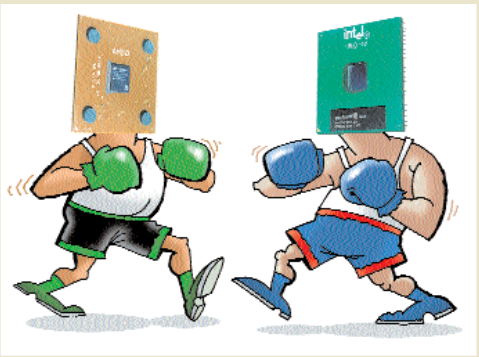


The GigaHertz Wars

The battle lines were drawn between AMD and Intel.
Round 1: **March 2000**
Athlon 1 GHz vs Pentium III 1 GHz
AMD released Athlon 1 GHz which had the L2 cache running at 333 MHz (L2 multiplier set at 1/3). Just two days after AMD's release came Intel's Pentium III 1 GHz, which had the L2 cache running



at full speed. The flipside however was that it made the Pentium III a nightmare to manufacture. AMD went for the kill and released the Thunderbird 1 GHz that had the L2 cache running at full core speed.
Round 2: **August 2001**
Pentium 4, 2 GHz vs Athlon XP
It took eight years to traverse 900 MHz (from 100 MHz to 1 GHz) and just over a year to reach the 2 GHz mark. The Pentium 4, 2 GHz is a testimony to the way the processor industry continually propels itself to new heights. Intel beat AMD to this landmark by an easy margin! Athlon XP languishes at 1.6 GHz.

was interesting to observe that the Pentium 4, 2 GHz could not beat the Athlon XP 1800+ despite having 466 MHz extra clock speed.
Even the 800 MHz processors will suffice for Office applications, unless you need to work with large chunks of data and heavy files (such as Photoshop files) on a regular basis. For the kind of applications and the kind of load that the average PC user puts the system through, the higher clock speeds are wasted.

Gaming: Where speed counts
What's the real reason behind processor manufacturers increasing the speed limits every few months? After all, how much difference does it make if Word opens a

second faster or slower? For the processors today, running these day-to-day applications is no sweat. Ask a gamer, and he'd probably look at you askance and reply: "Quake III".
Gaming and other entertainment applications are probably the only applications that will show a marked improvement in performance with incremental increases in processor performance and, as you will see in the test, the scores vary enormously with different processors and memory architectures.
Most games run processors at maximum throughput, pushing and trying to extract every single bit of performance. This also makes them great indicators of system stability and your system will most definitely crash if your processor has heated beyond its safety limit.
If you are a power gamer and want your gameplay to be of the highest quality with the fastest frame rates possible, you know where to look again: The Pentium 4, 2 GHz blazed ahead with a score of 314.1 fps in the normal mode benchmark of *Quake III Arena*, which is considered to have one of the most well-oiled gaming engines. The Pentium 4, 2 GHz takes gaming to a level where even if you've turned on all the razzle-dazzle and razzmatazz, you will still be able to play at a really fast frame rate. The Athlon XP 1800+ was beaten even by the Pentium 4, 1.7 GHz and came in third with a score of 276.4 fps. In the texture-heavy Direct3D game *Evolva*, things were slightly different and the Athlon XP 1800+ managed to edge ever so slightly ahead by scoring 148.2 as against the 144 fps logged by the Pentium 4, 2 GHz.
Another benchmark we ran was 3DMARK 2001, which is extremely performance sensitive and scales higher the minute you improve any of the subsystems. Gaining even an extra 100 marks in this test is very difficult. In this test,

the Athlon XP 1800+ scored 7060 as against the 6960 3DMARKS logged by the Pentium 4, 2 GHz.
These three gaming tests indicate that though the Athlon XP 1800+ will give you a good performance in games based on Direct3D (*Evolva* or *Max Payne*), you will get a better performance with the Pentium 4, 2 GHz when playing most games, especially those based on the *Quake III* engine, such as the newly released *Return to Castle Wolfenstein*.
If you are not too concerned about top-notch effects and are content with decent playable frame rates, consider the choices available in the performance category of processors. In this segment there was a discernible drop in performance in all the gaming tests. Nevertheless, the Athlon series held its own very well here, beating the Pentium III series in all the tests.
In the value segment there was an even further drop in performance—by as much as 100 fps in *Quake III* alone. In *Quake III Arena* the Duron 1.1 GHz logged a decent 196 fps in normal mode as against the 179.3 fps logged by the Celeron.
In this segment the 3DMARK 2001 scores dropped to under 6000 even though these processors were paired with the fastest graphics card around (nVidia GeForce3 64 MB DDR). If you have a casual interest in gaming then the Value segment should fulfil your requirements. However, keep in mind that using a high-end graphics card with such a processor would not serve any purpose.
The performance of the under 1 GHz processors in the gaming tests clearly shows that if you want an acceptable performance in most modern games, you'd be better off with a processor rated at 1 GHz or more.

Multimedia performance
When Intel introduced the Pentium MMX (Multimedia Extensions), it was for the first time that you had processors

